



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 28.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CS F317
Course Title : Reinforcement Learning
Instructor-in-Charge : Dr. Prashant Trivedi
Tutorial/Practical Instructors: NA

1. Course Description: Reinforcement Learning (RL) is based on the idea where we learn by interacting with our environment. It is a collection of machine learning techniques where agent(s) learn how to behave in an environment by performing actions and assessing the results. RL approach is building programs that learn how to predict and act in stochastic environment, based on experience. This course will provide an overview to students on some of the fundamental ideas on which modern RL is built including Markov decision processes, Multi-armed bandits value functions, Monte Carlo estimation, dynamic programming, TD methods, approximation methods, Actor-Critic methods, etc. This course will help students to understand and apply RL in several systems including video distribution systems, game development, IOT devices, robotics, clinical decision making, industrial process control, finance portfolio balancing, etc.

2. Scope and Objective of the Course:

This course aims to achieve the following goals:

- To provide students with the knowledge to structure a reinforcement learning problem.
- To introduce students to learn and apply basic RL algorithms for simple sequential decision-making problems in uncertain conditions
- To introduce students research and development work in reinforcement learning by interpreting state-of-the-art RL research and communicating their results.
- To provide knowledge to students to build a RL system that knows how to make automated decisions.
- To give students opportunities to understand the space of RL algorithms including Temporal Difference Learning, Monte Carlo, Q-Learning, approximation solution methods, A2C, A3C, etc.

3. Text Books:

1. Sutton, Richard S., and Andrew G. Barto. *Reinforcement learning: An introduction*. Second Edition, MIT press, 2018.

4. Reference Books:

1. Wiering, Marco A., and Martijn Van Otterlo. "Reinforcement learning." *Adaptation, learning, and optimization* 12.3 (2012).



5. Course Plan:

Lecture No.	Learning Outcomes	Topics to be covered	Reference
1-2	To Introduce Reinforcement Learning, its limitations and scope.	Introduction to Reinforcement Learning. A brief summary of the pre-requisites, mathematical and programming tools required.	Class notes
PART 1: Tabular Solution Methods			
3-4	To understand the application of RL for single-state Bandit problems.	Bandit problems	T1:Ch2
5-7	To introduce methods for solving finite markov decision problems	Finite Markov Decision Processes	T1:Ch3
8-9	To understand the utility of DP (require perfect model) in RL	Solution Methods: Dynamic Programming	T1: Ch4
10-12	To understand and apply Monte Carlo Methods (require only experience) for RL	Solution Methods: Monte Carlo Methods	T1:Ch5
13-18	To understand and apply the application of Temporal Difference Learning including Q-learning and Sarsa for RL	Solution Methods: Temporal-Difference Learning	T1:Ch6, Ch7 and Class notes
19-22	To develop a unified view of methods that require a model of the environment and methods that can be used without a model.	Planning and Learning with Tabular Methods	T1:Ch8
PART 2: Approximate Solution Methods			
23-28	To apply RL for applications where most states encountered will never have been experienced exactly before. To apply on-policy and off-policy approximation of action values.	Function approximation for generalization	T1:Ch9-Ch11 and Class Notes
29-34	To understand and learn the reinforcement learning in a multiagent setting	Multi-agent reinforcement learning	Class Notes



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35-40	To understand the application of A2C and A3C (with multiple independent agents). RLHF	Actor-Critic (A2C, A3C), RLHF	Class Notes
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6. Evaluation Scheme:

Component	Duration (Minutes)	Weightage	Date & Time	Nature
Mid-semester Sem	90 Mins	20%	TBA	Closed Book
Course Project (with final presentation and viva)	-	35% (10% will be evaluated before the mid-sem)	Details will be announced during the lecture sessions.	Open
Assignment/Quiz	-	10%	Quizzes will be taken during the lecture hours. Details will be announced in the class.	Closed
Comprehensive Examination	180 Mins	35%	13/12/2026	Closed Book + Open Book

7. Pre-requisites: A course on Linear Algebra, Probability and Statistics. Coding in Python.

Background in ML or Deep Learning is a plus but not required.

8. Chamber Consultation Hour: Consultation hours will be announced in the class.

9. Notices: All notices pertaining to this course will be displayed on the Moodle.

10. Make-up Policy: The make-up policy will be in accordance with the AUGSD guidelines.

11. Attendance Policy: Compulsory in first week only.

Instructor-in-charge
Course No. CS F317